

EGFR Mutation Promotes Glioblastoma through Epigenome and Transcription Factor Network Remodeling.

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Epidermal growth factor receptor (EGFR) gene amplification and mutations are the most common oncogenic events in glioblastoma (GBM), but the mechanisms by which they promote aggressive tumor growth are not well understood. Here, through integrated epigenome and transcriptome analyses of cell lines, genotyped clinical samples, and TCGA data, we show that EGFR mutations remodel the activated enhancer landscape of GBM, promoting tumorigenesis through a SOX9 and FOXG1-dependent transcriptional regulatory network in vitro and in vivo. The most common EGFR mutation, EGFRvIII, sensitizes GBM cells to the BET-bromodomain inhibitor JQ1 in a SOX9, FOXG1-dependent manner. These results identify the role of transcriptional/epigenetic remodeling in EGFR-dependent pathogenesis and suggest a mechanistic basis for epigenetic therapy.

The EGFRvIII transcriptome in glioblastoma: A meta-omics analysis.

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EGFR is among the genes most frequently altered in glioblastoma, with exons 2-7 deletions (EGFRvIII) being among its most common genomic mutations. There are conflicting reports about its prognostic role and it remains unclear whether and how it differs in signaling compared with wildtype EGFR. To better understand the oncogenic role of EGFRvIII, we leveraged 4 large datasets into 1 large glioblastoma transcriptome dataset (n = 741) alongside 81 whole-genome samples from 2 datasets. The EGFRvIII/EGFR expression ratios differ strongly between tumors and range from 1% to 95%. Interestingly, the slope of relative EGFRvIII expression is near-linear, which argues against a more positive selection pressure than EGFR wildtype. An absence of selection pressure is also suggested by the similar survival between EGFRvIII-positive and -negative glioblastoma patients. EGFRvIII levels are inversely correlated with pan-EGFR (all wildtype and mutant variants) expression, which indicates that EGFRvIII has a higher potency in downstream pathway activation. EGFRvIII-positive glioblastomas have a lower CDK4 or MDM2 amplification incidence than EGFRvIII-negative (P = .007), which may point toward crosstalk between these pathways. EGFRvIII-expressing tumors have an upregulation of "classical" subtype genes compared to those with EGFR-amplification only (P = 3.873e-6). Genomic breakpoints of the EGFRvIII deletions have a preference toward the 3'-end of the large intron-1. These preferred breakpoints preserve a cryptic exon resulting in a novel EGFRvIII variant and preserve an intronic enhancer. These data provide deeper insights into the complex EGFRvIII biology and provide new insights for targeting EGFRvIII mutated tumors.

LRIG1 negatively regulates the oncogenic EGF receptor mutant EGFRvIII.

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Epidermal growth factor receptor (EGFR) mutation is frequently observed in human cancer and contributes to the growth, survival and therapeutic resistance of tumors. EGFRvIII is an oncogenic EGFR mutant resulting from the deletion of exons 2-7 and is the most common EGFR mutant observed in glioblastoma multiforme, an aggressive brain tumor. EGFRvIII is constitutively active but poorly

ubiquitinated, leading to inefficient receptor trafficking to lysosomes and unattenuated oncogenic signaling. The mechanism by which EGFRvIII evades downregulation is not fully understood although recent studies suggest that its interaction with the ubiquitin ligase Cbl may be compromised. In this study, we examine the regulation of EGFRvIII by the recently identified negative regulator, LRIG1, which targets EGFR through recognition of its extracellular domain. Here, we determine whether the extracellular domain deletion in EGFRvIII renders it refractory to LRIG1 regulation. We find that EGFRvIII retains interaction with LRIG1 and is in fact more sensitive to LRIG1 action than wild-type receptor. We demonstrate that LRIG1 regulation of EGFRvIII is distinct from the only other known mechanism of EGFR regulation, Cbl-mediated degradation. Ectopic expression of LRIG1 in EGFRvIII(+) glioblastoma cells opposes EGFRvIII-driven tumor cell proliferation, survival, motility and invasion. Finally, RNAi-mediated silencing of LRIG1 alters EGFRvIII intracellular trafficking and leads to enhanced EGFRvIII expression, suggesting that loss of LRIG1 in tumors may contribute to a permissive environment for EGFRvIII overexpression, contributing to EGFRvIII oncogenesis.

The effect of epidermal growth factor receptor variant III on glioma cell migration by stimulating ERK phosphorylation through the focal adhesion kinase signaling pathway.

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Epidermal growth factor receptor variant III (EGFRvIII), the most common EGFR mutation, is associated with cell migration of glioblastoma multiforme (GBM) cases; however, the mechanism has not been elucidated. In this study, we found that the EGFRvIII-promoted glioma cell migration was closely linked to high levels of tyrosine phosphorylation in focal adhesion kinase (FAK) Y397. We also demonstrated that EGFRvIII formed a complex with FAK, resulting in enhanced tyrosine phosphorylation levels of FAK Y397 and EGFR Y1068. After knockdown of FAK expression via anti-FAK shRNA, the U87ΔEGFR cell migration was significantly inhibited, accompanying with the reduced phosphorylation levels of extracellular signal-regulated kinase (ERK1/2). Furthermore, the role of ERK1/2 in FAK-regulated cell migration was confirmed. Taken together, our results suggest that FAK and its downstream molecule ERK were involved in EGFRvIII-promoted glioma cell migration in U87ΔEGFR cells.

Epidermal growth factor receptor variant type III markedly accelerates angiogenesis and tumor growth via inducing c-myc mediated angiopoietin-like 4 expression in malignant glioma.

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Expression of the constitutively activated mutant EGFR variant III (EGFRvIII), the most common mutation in glioblastoma multiforme (GBMs), has been clinically correlated with tumor proliferation, invasion, and angiogenesis. In this study, we examined the role of EGFRvIII on the tumor microenvironment, especially on angiogenesis. To study the role of EGFRvIII in tumor angiogenesis, we prepared LN229 glioblastoma transfected with enhanced green fluorescent protein (EGFP), wild-type EGFR, or EGFRvIII (LN229-WT or -vIII), and examined tumor growth and microvessel density in the tumors. Additionally, the potential angiogenic factors were identified by real-time PCR analysis, and the functions in LN229-vIII cells were examined. LN229-vIII cells showed more aggressive tumor growth and higher vascularity as compared to LN229-WT cells in vivo, although there was no significant difference in the cell growth rates in vitro. We

next investigated the expression of 60 angiogenesis-related factors to clarify the mechanisms underlying the difference in vascularity between tumor xenografts of LN229-vIII and LN229-WT. We found that the mRNA and protein expressions of angiopoietin-like 4 (Angptl4), a secreted protein involved in angiogenesis and metabolism regulation, were significantly induced by EGFRvIII overexpression, both in vitro and in vivo. Constitutive knockdown of Angptl4 in LN229-vIII using shRNA significantly decreased the microvessel density in the tumor xenografts and suppressed tumor growth. To clarify the regulatory mechanisms of Angptl4 by EGFRvIII, we analyzed the signaling pathways and transcription factors by pharmacological inhibition and RNA interference. U0126, an ERK signal inhibitor dramatically suppressed Angptl4 expression. The transcription factor c-Myc, which is regulated by ERK, was activated in the LN229-vIII cells and knockdown of c-Myc using siRNA also attenuated Angptl4 expression in the LN229-vIII cells. Furthermore, chromatin immunoprecipitation (ChIP) assay revealed increased recruitment of c-Myc to the promoter region of Angptl4 in the LN229-vIII cells. In summary, we demonstrated that EGFRvIII induces Angptl4 expression through the ERK/c-Myc pathway and promotes tumor angiogenesis in malignant gliomas.

ERK-mediated negative feedback regulation of oncogenic EGFRvIII in glioblastoma cells.

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Epidermal growth factor receptor variant III (EGFRvIII) is the most common active EGFR mutant in glioblastoma multiforme (GBM). The expression of this mutant often correlates with a poor patient prognosis due to its ability to extend downstream signaling. The EGFR pathway is controlled by a negative feedback mechanism that restricts the extent and length of downstream signaling. To date, the role of negative feedback in the oncogenic EGFRvIII mutant remains undetermined. The present study indicated that activation of the MEK-ERK pathway led to the phosphorylation of Thr-402, a conserved negative feedback residue in the juxtamembrane domain corresponding to Thr-669 of wild-type EGFR (EGFRwt), which resulted in a rapid reduction in the tyrosine phosphorylation of EGFRvIII in U87MG human glioblastoma and 293 cells. Moreover, despite the incapability of EGFRvIII to bind ligands, EGF was indicated to downregulate the tyrosine phosphorylation of EGFRvIII by activating the EGFRwt-ERK pathway. These results demonstrated a conserved negative feedback mechanism in the activation of EGFRvIII, which presents a new aspect in functional interactions between EGFRvIII and EGFRwt in glioblastoma cells.

Tumor Suppressing Subtransferable Candidate 4 Expression Prevents Autophagy-Induced Cell Death Following Temozolomide Treatment in Glioblastoma Cells.

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Glioblastoma (GBM) is the most common and aggressive type of brain cancer in adults, with temozolomide (TMZ) being widely used as the standard chemotherapy drug for its treatment. However, GBM frequently becomes resistant to TMZ treatment due to various mechanisms including amplification and mutations of the epidermal growth factor receptor (EGFR), where EGFR variant III (EGFRvIII) is the most common EGFR mutation. Autophagy (macroautophagy) is an intracellular "self-degradation" process involving the lysosome. It mainly plays a pro-cell survival role contributing to drug resistance in

cancers including GBM, but, under some conditions, it can induce cell death called autophagy-induced cell death (AulCD). We recently published that TSSC4 (tumor suppressing subtransferable candidate 4) is a novel tumor suppressor and a novel autophagy inhibitor that inhibits cancer cell growth through its interacting with the autophagy protein LC3. In this brief research report, we demonstrate that cell death induced by TMZ in GBM cells is inhibited by overexpression of TSSC4. TSSC4 overexpression also prevents TMZ-induced autophagy but not when TSSC4 is mutated in its conserved LC3-interacting region. When EGFRvIII was expressed in GBM cells, TSSC4 protein was increased and TMZ-induced cell death was decreased. Knockout of TSSC4 in EGFRvIII-expressing GBM cells increased TMZ-induced autophagy and cell death. This cell death was decreased by autophagy inhibition, suggesting that TSSC4 downregulation promotes TMZ-induced AulCD. This indicates that TSSC4 is a novel target to sensitize GBM cells to TMZ treatment.

EGFRvIII Confers Sensitivity to Saracatinib in a STAT5-Dependent Manner in Glioblastoma.

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Glioblastoma (GBM) is the most common primary malignant brain tumor in adults, with few effective treatments. EGFR alterations, including expression of the truncated variant EGFRvIII, are among the most frequent genomic changes in these tumors. EGFRvIII is known to preferentially signal through STAT5 for oncogenic activation in GBM, yet targeting EGFRvIII has yielded limited clinical success to date. In this study, we employed patient-derived xenograft (PDX) models expressing EGFRvIII to determine the key points of therapeutic vulnerability within the EGFRvIII-STAT5 signaling axis in GBM. Our findings reveal that exogenous expression of paralogs STAT5A and STAT5B augments cell proliferation and that inhibition of STAT5 phosphorylation in vivo improves overall survival in combination with temozolomide (TMZ). STAT5 phosphorylation is independent of JAK1 and JAK2 signaling, instead requiring Src family kinase (SFK) activity. Saracatinib, an SFK inhibitor, attenuates phosphorylation of STAT5 and preferentially sensitizes EGFRvIII+ GBM cells to undergo apoptotic cell death relative to wild-type EGFR. Constitutively active STAT5A or STAT5B mitigates saracatinib sensitivity in EGFRvIII+ cells. In vivo, saracatinib treatment decreased survival in mice bearing EGFR WT tumors compared to the control, yet in EGFRvIII+ tumors, treatment with saracatinib in combination with TMZ preferentially improves survival.

Prognostic effect of epidermal growth factor receptor and EGFRvIII in glioblastoma multiforme patients.

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The epidermal growth factor receptor (EGFR) is overexpressed in approximately 50% to 60% of glioblastoma multiforme tumors, and the most common EGFR mutant, EGFRvIII, is expressed in 24% to 67% of cases. We sought to determine whether glioblastoma multiforme expression of either overexpressed wild-type EGFR or the mutant EGFRvIII is an independent predictor of overall patient survival. Glioblastoma multiforme patients (n = 196) underwent a > or =95% volumetric tumor resection followed by conformal radiation. Their EGFR and EGFRvIII status was determined by immunohistochemistry and survival analyses were done. In our study of glioblastoma multiforme patients, 46% (n = 91) failed to express EGFR, 54% (n = 105) had overexpression of the wild-type EGFR,

and 31% (n = 61) also expressed the EGFRvIII. Patients within groups expressing the EGFR, EGFRvIII, or lacking EGFR expression did not differ in age, sex, Karnofsky performance scale score, extent of tumor resection, or radiation. The median overall survival times for patients with tumors having EGFR expression absent, overexpressed only, or mutant (EGFRvIII) were 0.96, 0.98, and 1.07 years, respectively. However, for patients surviving > or =1 year, these values were 2.03, 2.02, and 1.21 years (P or =1 year, the expression of EGFRvIII was an independent negative prognostic indicator.

EGFRvIII gene rearrangement is an early event in glioblastoma tumorigenesis and expression defines a hierarchy modulated by epigenetic mechanisms.

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Amplification and rearrangements of the epidermal growth factor receptor (EGFR) gene are frequently found in glioblastoma multiforme (GBM). The most common variant is EGFR variant III (EGFRvIII). Research suggests that EGFRvIII could be a marker for a cancer stem cell or tumor-initiating population. If amplification and rearrangement are early events in tumorigenesis, this implies that they should be preserved throughout the tumor. However, in primary GBM, EGFRvIII expression is focal and sporadic. Unexpectedly, we found EGFR amplification and rearrangement throughout the tumor, including regions with no EGFRvIII expression, suggesting that mechanisms exist to modulate EGFRvIII expression even in the presence of high gene amplification. To study this phenomenon, we characterized three GBM cell lines with endogenous EGFRvIII. EGFRvIII expression was heterogeneous, with both positive and negative populations maintaining the genetic alterations, akin to primary tumors. Furthermore, EGFRvIII defined a hierarchy where EGFRvIII-positive cells gave rise to additional positive and negative cells. Only cells that had recently lost EGFRvIII expression could re-express EGFRvIII, providing an important buffer for maintaining EGFRvIII-positive cell numbers. Epigenetic mechanisms had a role in maintaining heterogeneous EGFRvIII expression. Demethylation induced a 20-60% increase in the percentage of EGFRvIII-positive cells, indicating that some cells could re-express EGFRvIII. Surprisingly, inhibition of histone deacetylation resulted in a 50-80% reduction in EGFRvIII expression. Collectively, this data demonstrates that EGFR amplification and rearrangement are early events in tumorigenesis and EGFRvIII follows a model of hierarchical expression. Furthermore, EGFRvIII expression is restricted by epigenetic mechanisms, suggesting that drugs that modulate the epigenome might be used successfully in glioblastoma tumors.